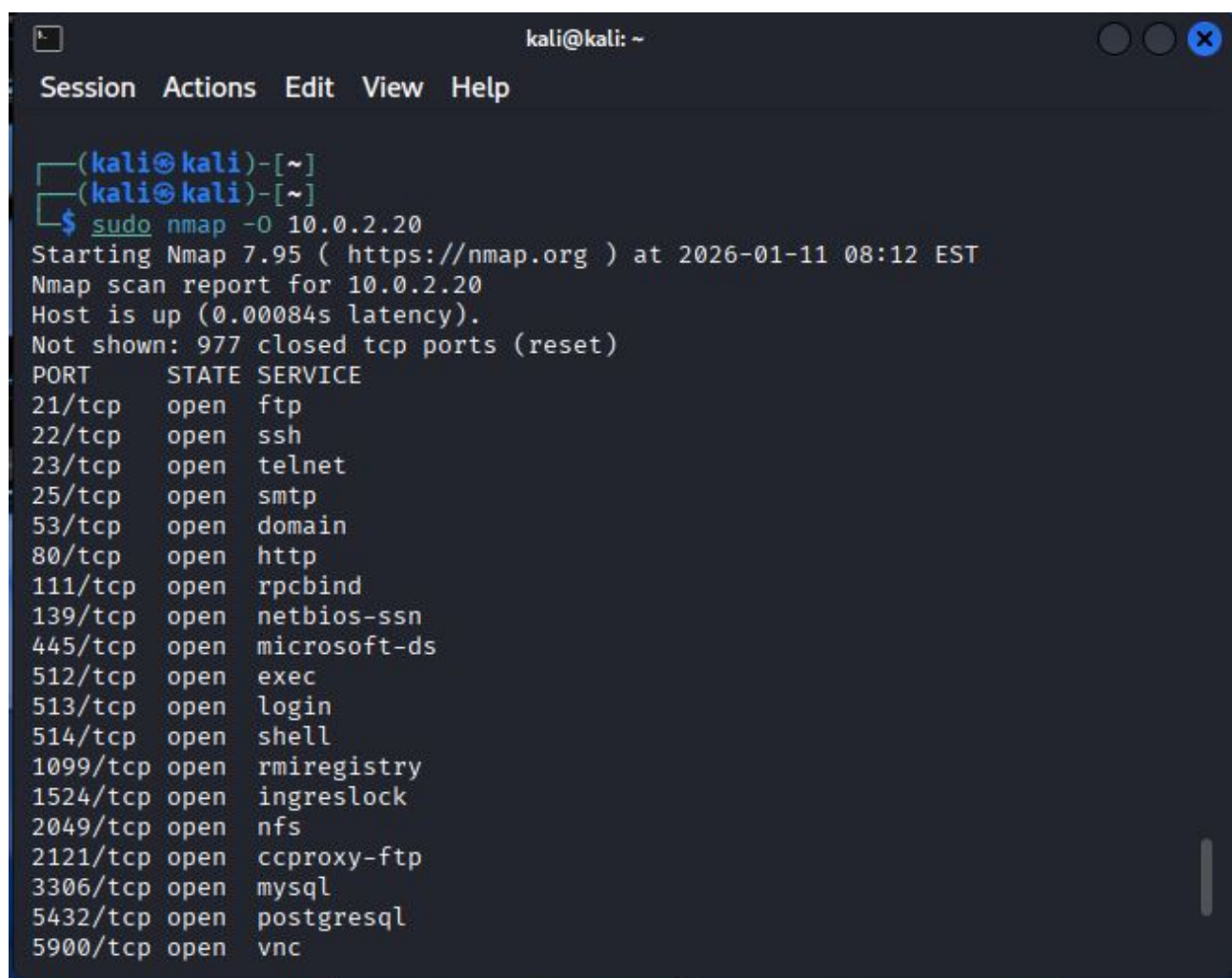


ESERCIZIO: Scansione dei servizi con Nmap

Prima di avviare le scansioni con Nmap ho effettuato la configurazione delle rete virtuale assicurandomi che le tre macchine virtuali (kali, meta, windows) fossero nella stessa RETE NAT.

Subito dopo è stato eseguito un comando `sudo nmap -O 10.0.2.20` per effettuare una scansione al fine di indentificare il sistema operativo di Metasploitable utilizzando Nmap. Il risultato indica che la macchina virtuale utilizza un sistema operativo LINUX.



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
(kali@kali)-[~]  
$ sudo nmap -O 10.0.2.20  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-11 08:12 EST  
Nmap scan report for 10.0.2.20  
Host is up (0.00084s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc
```

```
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:3A:6B:6B (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.08 seconds

(kali@kali)-[~]
$
```

Ho effettuato, quindi, una seconda scansione (SYN SCAN) per individuare la porte TCP aperte con il comando

Ho effettuato il comando: `sudo nmap -sS 10.0.2.20`

```
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo nmap -sS 10.0.2.20
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-11 08:15 EST
Nmap scan report for 10.0.2.20
Host is up (0.00042s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
```

```
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:3A:6B:6B (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 0.55 seconds

(kali@kali)-[~]
$
```

La scansione ha permesso di identificare i servizi in ascolto sul sistema

Come terzo passaggio ho effettuato una scansione TCP connect scan per verificare le porte aperte tramite una connessione completa con il comando `sudo nmap -sT 10.0.2.20`

```
kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ sudo nmap -sT 10.0.2.20
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-11 08:18 EST
Nmap scan report for 10.0.2.20
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
```



```
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:3A:6B:6B (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
```

```
Nmap done: 1 IP address (1 host up) scanned in 0.52 seconds
```

```
(kali@kali)-[~]  
$
```

Questo tipo di scansione completa l'handshake TCP ed è più facilmente rilevabile rispetto alla SYN scan.

Dopo aver individuato le porte aperte ho cercato di identificare quali servizi fossero in esecuzione e le relative versioni con il comando `sudo nmap -sV 10.0.2.20`

```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 0.52 seconds  
  
(kali@kali)-[~]  
$ sudo nmap -sV 10.0.2.20  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-11 08:25 EST  
Nmap scan report for 10.0.2.20  
Host is up (0.00070s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rshcd  
513/tcp   open  login?  
514/tcp   open  shell        Netkit rshd  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)
```

```

6000/tcp open  X11          (access denied)
6667/tcp open  irc          UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:3A:6B:6B (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs
: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 54.03 seconds

(kali@kali)-[~]
$

```

Dall'analisi dei dati forniti da NMAP con l'HOST is up ho verificato che fosse attivo, da PORT STATE ho individuato numerose porte TCP aperte. Analizzando la colonna SERVICE ho notato che sulle porte aperte sono presenti diversi servizi di rete attivi. Infine, analizzando la colonna VERSION, NMAP ha indentificato le versioni dei principali sistemi in esecuzione evidenziando la presenza di software datato.

Effettuato lo stesso test sulla macchina virtuale windows

Comando: `sudo nmap -O 10.0.2.3` per risalire al sistema operativo

```

(kali@kali)-[~]
$ sudo nmap -O 10.0.2.3
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-11 08:51 EST
Nmap scan report for 10.0.2.3
Host is up (0.0020s latency).
Not shown: 981 closed tcp ports (reset)
PORT      STATE SERVICE
7/tcp     open  echo
9/tcp     open  discard
13/tcp    open  daytime
17/tcp    open  qotd
19/tcp    open  chargen
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
1801/tcp  open  msmq
2103/tcp  open  zephyr-clt
2105/tcp  open  eklogin
2107/tcp  open  msmq-mgmt
3389/tcp  open  ms-wbt-server
5357/tcp  open  wsdapi
5432/tcp  open  postgresql
8009/tcp  open  ajp13
8080/tcp  open  http-proxy

```

```
8080/tcp open  http-proxy
8443/tcp open  https-alt
MAC Address: 08:00:27:E6:3A:C0 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1507 - 1607
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.36 seconds

(kali@kali)-[~]
$
```

Comando: `sudo nmap -sS 10.0.2.3` scansione (SYN SCAN) per individuare la porte TCP

```
(kali@kali)-[~]
$ sudo nmap -sS 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-11 09:03 EST
Nmap scan report for 10.0.2.3
Host is up (0.0016s latency).
Not shown: 981 closed tcp ports (reset)
PORT      STATE SERVICE
7/tcp     open  echo
9/tcp     open  discard
13/tcp    open  daytime
17/tcp    open  qotd
19/tcp    open  chargen
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
1801/tcp  open  msmq
2103/tcp  open  zephyr-clt
2105/tcp  open  eklogin
2107/tcp  open  msmq-mgmt
3389/tcp  open  ms-wbt-server
5357/tcp  open  wsdapi
5432/tcp  open  postgresql
8009/tcp  open  ajp13
8080/tcp  open  http-proxy
8443/tcp  open  https-alt
MAC Address: 08:00:27:E6:3A:C0 (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)

Nmap done: 1 IP address (1 host up) scanned in 2.80 seconds

(kali@kali)-[~]
$
```


Comando `sudo nmap -sT 10.0.2.3` scansione completa del TCP e più facilmente rilevabile rispetto alla SYN scan.

```
(kali@kali)-[~]
$ sudo nmap -sT 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-11 09:09 EST
Nmap scan report for 10.0.2.3
Host is up (0.00085s latency).
Not shown: 981 closed tcp ports (conn-refused)
PORT      STATE SERVICE
7/tcp     open  echo
9/tcp     open  discard
13/tcp    open  daytime
17/tcp    open  qotd
19/tcp    open  chargen
80/tcp    open  http
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
1801/tcp  open  msmq
2103/tcp  open  zephyr-clt
2105/tcp  open  eklogin
2107/tcp  open  msmq-mgmt
3389/tcp  open  ms-wbt-server
5357/tcp  open  wsapi
5432/tcp  open  postgresql
8009/tcp  open  ajp13
8080/tcp  open  http-proxy
8443/tcp  open  https-alt
MAC Address: 08:00:27:E6:3A:C0 (PCS Systemtechnik/Oracle VirtualBox virtual N
```

Comando `sudo nmap -sV 10.0.2.3` per identificare quali servizi sono in esecuzione e le relative versioni

```
(kali@kali)-[~]
$ sudo nmap -sV 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-11 09:11 EST
Stats: 0:01:11 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 89.47% done; ETC: 09:12 (0:00:08 remaining)
Stats: 0:02:09 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 94.74% done; ETC: 09:13 (0:00:07 remaining)
Stats: 0:02:14 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 94.74% done; ETC: 09:13 (0:00:07 remaining)
Nmap scan report for 10.0.2.3
Host is up (0.00067s latency).
Not shown: 981 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
7/tcp     open  echo
9/tcp     open  discard?
13/tcp    open  daytime         Microsoft Windows International daytime
17/tcp    open  qotd            Windows qotd (English)
19/tcp    open  chargen
80/tcp    open  http            Microsoft IIS httpd 10.0
135/tcp   open  msrpc           Microsoft Windows RPC
139/tcp   open  netbios-ssn     Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds    Microsoft Windows 7 - 10 microsoft-ds (workgroup
: WORKGROUP)
1801/tcp  open  msmq?
2103/tcp  open  msrpc           Microsoft Windows RPC
2105/tcp  open  msrpc           Microsoft Windows RPC
```

```
2107/tcp open  msrpc           Microsoft Windows RPC
3389/tcp open  ms-wbt-server   Microsoft Terminal Services
5357/tcp open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5432/tcp open  postgresql?
8009/tcp open  ajp13           Apache Jserv (Protocol v1.3)
8080/tcp open  http            Apache Tomcat/Coyote JSP engine 1.1
8443/tcp open  ssl/https-alt
MAC Address: 08:00:27:E6:3A:C0 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Host: DESKTOP-9K104BT; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 160.46 seconds

(kali@kali)-[~]
$
```

CONCLUSIONE

La scansione ha permesso di individuare le porte aperte, i servizi in ascolto e le relative versioni su META e WINDOWS. I risultati mostrano la presenza di numerosi servizi attivi e versioni software vecchie, confermando la vulnerabilità del sistema analizzato.